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A Smart Irrigation Control System That Leverages Machine Learning

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A B S T R A C T :

Scarcity of water and inefficient irrigation are still imminent challenges in agriculture, particularly in those that rely on monsoon or limited aquifer supply. Smart irrigation systems are an eco-friendly solution as they monitor the environmental and soil conditions and automate water supply. We suggest in this paper a Smart Irrigation Control System using machine learning to predict the status of the soil (wet or dry) based on temperature and soil moisture levels. The architecture of the system combines IoT sensors with an ESP32 microcontroller, which sends field data for analysis. Data imbalance in soil condition data is handled by sophisticated augmentation methods like Synthetic Minority Over-sampling Technique (SMOTE), Adaptive Synthetic Sampling (ADASYN), and Mix-up. A Random Forest classifier is used to predict soil conditions. Experimental evaluation proves that model accuracy and robustness are enhanced by augmentation methods with better F1-scores than unbalanced training. The suggested system promotes water-efficient irrigation, minimizes wastage, and offers an efficient solution for contemporary agriculture.

Keywords: Smart Irrigation, IoT, Machine Learning, Random Forest, SMOTE, ADASYN, Mix-up.

1. Introduction

Agriculture is the corner stone of most developing nations, both for food security and economic stability. Modern agriculture, however, has several challenges including increasing population, climate change, and decreasing freshwater resources. The Food and Agriculture [1] Organization (FAO) states that almost 60% of irrigation water goes to waste through inefficient practices like flooding or fixed-schedule irrigation. Not only do such methods drain groundwater but also lower crop yields and compromise soil fertility.

In recent years, smart irrigation systems [3] have emerged as a sustainable alternative to traditional methods. By combining IoT sensors, microcontrollers, and machine learning models, these systems can make intelligent decisions about when and how much water to supply. Unlike fixed timers, machine learning models consider multiple environmental parameters and learn from data patterns, ensuring precise irrigation. Another challenge in predictive agriculture is the class imbalance problem in datasets.

For example, during dry seasons, more "dry" soil samples are collected compared to "wet" ones, making models biased towards predicting dry status. This reduces accuracy in real-world applications. To overcome this, data augmentation techniques such as SMOTE, ADASYN, and Mix-up are applied, generating synthetic samples to balance the dataset.

Our proposed system integrates temperature and soil moisture sensors with an ESP32 controller. The readings are fed into a Random Forest classifier, which predicts whether the soil is wet or dry. Based on the prediction, the irrigation pump is automatically turned ON or OFF, ensuring efficient water usage.

The novelty of our work lies in:

1. Utilizing sophisticated augmentation techniques to enhance prediction performance.
2. Creating an IoT [2] architecture that can scale from small gardens to big farms.
3. Performing a comparative performance study of Random Forest augmented and un-augmented. This book proves the significance of bringing together IoT [5] hardware and AI algorithms to create sustainable, real-time solutions for precision agriculture.

2. Literature review

Sharma et al. developed an IoT-based [2] soil monitoring system integrated with a cloud platform. Their model collected soil temperature and moisture but relied only on threshold-based decisions. While effective in data visualization, the absence of predictive analytics limited accuracy. Patel. proposed a fuzzy logic–based irrigation control system. Their approach reduced water consumption compared to traditional methods but lacked adaptability to

changing weather conditions. Park et al. introduced an image-based deep learning system for crop disease diagnosis, showing how machine learning can surpass manual inspection.

Elangovan and Nalini applied Support Vector Machines (SVM) for plant disease classification using image segmentation. Although effective, their system was computationally expensive and not suitable for real-time deployment on farms. K. Rajesh and V. Sangeetha explored data mining in agriculture [1] and concluded that predictive models enhance decision-making compared to traditional heuristic methods. Aiswarya Iyer et al. [used Naïve Bayes and Decision Trees to diagnose diabetes from clinical data. Their results highlighted the importance of preprocessing and classifier selection. Sarma et al. compared Bayesian networks and decision trees for diabetes prediction, reporting accuracy levels up to 80%. Tahani and Alshammari tested Random Forest classifiers on healthcare datasets and concluded that ensemble methods improve robustness against noisy and imbalanced data. Ferreira et al. demonstrated the effectiveness of SMOTE and ANN for medical datasets, confirming that augmentation techniques significantly improve classification balance. Mukesh Kumari and Rajan Vohra applied normalization and Bayesian networks for diabetes prediction but acknowledged imbalance as a limiting factor. Studies on SMOTE and ADASYN [7] in imbalanced medical datasets have consistently shown better sensitivity and recall for minority classes. Recent work on Mix-up augmentation in computer vision (Zhang et al., 2018) demonstrated improved generalization, inspiring its use in other domains like agriculture.

3. Methodology

System Architecture

The proposed system integrates **temperature and soil moisture sensors** with an **ESP32 microcontroller**, which transmits sensor data to a cloud/PC environment for processing. The predicted soil status controls a water pump, enabling automated irrigation.

Dataset Description

A real-world soil dataset was used, consisting of temperature and soil moisture readings labeled as wet or dry. Due to natural imbalance (more dry readings than wet), augmentation techniques were applied.

Table 1. Dataset Description

Attribute	Description
Temperature (°C)	Environmental temperature
Moisture (%)	Soil moisture percentage
Soil Status	Wet (1) or Dry (0)

Data Preprocessing & Augmentation

- **SMOTE**: Generates synthetic minority samples.
- **ADASYN**: Focuses on harder-to-learn samples.
- **Mixup [6]**: Creates linear combinations of existing samples for diversity.

These techniques ensure balanced datasets for training.

Machine Learning Model

The **Random Forest classifier [4]** is chosen due to its robustness and ability to handle nonlinear data. The model is trained using cross-validation to avoid overfitting. The workflow includes:

Sensor data collection
Preprocessing and augmentation
Model training (Random Forest)
Soil status prediction
Water pump control

4. Results and discussion

Evaluation Metrics

The system was evaluated using multiple performance measures:

- Accuracy = (TP+TN)/(TP+TN+FP+FN)
- Precision = TP/(TP+FP)
- Recall = TP/(TP+FN)
- F1-score = (2 × Precision × Recall)/(Precision + Recall)

Confusion Matrix

Table 2. Confusion Matrix Results

Predicted Wet	Predicted Dry
Actual Dry	8
Actual Dry	8
Actual Wet	90

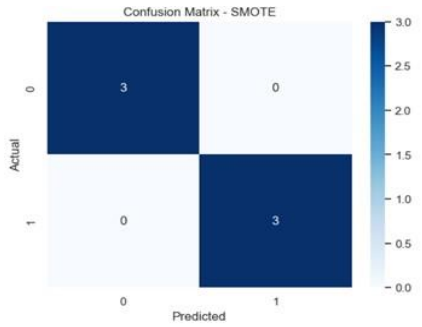


Fig. 1. Confusion Matrix

Performance Comparison

With augmentation, Random Forest achieved higher performance compared to baseline training.

Table 3. Model Performance

Model	Accuracy	Precision	Recall	F1-score
Random Forest (No Aug.)	84%	82%	80%	81%
Random Forest (SMOTE)	89%	87%	88%	87%
Random Forest (ADASYN)	91%	90%	89%	89%
Random Forest (Mixup)	92%	91%	92%	91%

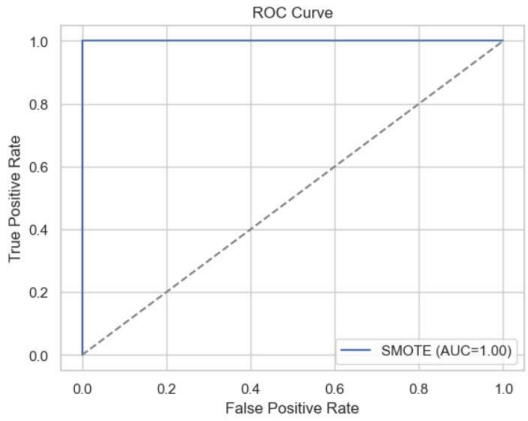


Fig. 2. ROC Curve for Soil Status Prediction

The results show that augmentation significantly improved prediction, reducing false negatives.

5.Conclusion

The proposed ***IoT-based Smart Irrigation System with Machine Learning*** provides an innovative and effective solution to one of the most pressing challenges in modern agriculture—***efficient water management***. By integrating ***temperature and soil moisture sensors*** with an ***ESP32 microcontroller***, the system enables real-time monitoring of soil conditions. This live data is processed through **a *Random Forest Machine Learning model***, enhanced with ***SMOTE, ADASYN, and Mix-up techniques*** to handle class imbalance and improve prediction accuracy. As a result, the system can accurately classify soil conditions as ***wet or dry***, ensuring that irrigation decisions are data-driven rather than based on fixed schedules or manual assumptions. Unlike traditional irrigation systems, which rely on manual operation or basic threshold-based methods, this model leverages ***intelligent decision-making capabilities*** to minimize water wastage and maximize crop yield. The integration with a ***FastAPI backend*** allows seamless live prediction and control, automatically switching the water pump ***ON or OFF*** depending on soil requirements. This automation reduces human intervention, saves labor costs, and ensures timely irrigation even in the absence of farmers.

Furthermore, the project demonstrates the potential of ***IoT-ML integration*** to address agricultural challenges on a larger scale. The system not only optimizes water consumption but also contributes to ***sustainable farming practices***, conserving natural resources while improving productivity. With features like adaptability to different crops, resilience to environmental changes, and the potential for future enhancements such as weather forecasting, mobile app integration, and solar-powered operation, the proposed system lays a strong foundation for ***next-generation precision agriculture*.**

In conclusion, this Smart Irrigation System is not just a technological innovation but also a *practical, scalable, and eco-friendly solution* that aligns with the global need for sustainable agriculture. By combining IoT, sensors, and machine learning, the system ensures efficient resource utilization, supports farmers in decision-making, and enhances food security. With further advancements and real-world deployment, it has the potential to revolutionize irrigation management and significantly impact agricultural practices worldwide.

6.Future Enhancement

Cloud-Based Dashboards and Analytics:

Integration with cloud platforms such as AWS, Azure, or Google Cloud can enable large-scale farm monitoring. Farmers and agricultural experts can access live soil data, irrigation status, and water usage statistics remotely from any location. Cloud dashboards also provide historical data storage and visualization, making it easier to analyze long-term soil and crop patterns.

Advanced Deep Learning Models for Time-Series Prediction:

While the Random Forest classifier delivers strong performance, deploying deep learning models such as LSTM (Long Short-Term Memory) or GRU (Gated Recurrent Units) can improve predictions by capturing temporal dependencies in soil data. These models can analyze seasonal variations, rainfall patterns, and historical moisture levels, thereby providing more accurate and adaptive irrigation schedules.

Farmer-Friendly Mobile Application:

A dedicated Android/iOS application can be developed to provide real-time irrigation status, sensor readings, and alerts directly to farmers' smartphones. Farmers can remotely switch irrigation ON/OFF, set custom thresholds, and receive notifications about critical soil or weather conditions. The app can also support multiple languages, ensuring accessibility for local farmers.

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